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## **7CUSMNDA Network Data Analysis Coursework**

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Github:

<https://github.com/qwe12345113/7CUSMNDA-Network-Data-Analysis->

## Part 1 - Task A: Network construction

### Question 1: How were the three Wikidata editor networks constructed?

To construct the Wikidata editor networks, three datasets representing small-, medium-, and large-scale discussion environments were selected: `BOT_REQUESTS`, `PROPERTIES`, and `PROJECT_CHAT`. In each dataset, editors were represented as nodes, and an edge was established between two editors whenever both contributed to the same page-thread discussion group. The resulting networks were modelled as weighted undirected graphs, where edge weights record the number of distinct discussion groups shared by each pair of editors. This specification is appropriate because the available data capture co-participation within discussion contexts, but do not reliably encode directed reply relations between individual editors.

Following the removal of missing values and exact duplicate records, the data were grouped by page and thread so that each group corresponded to a single discussion context. Within each group, unique participating editors were identified and all unordered editor pairs were generated and added to the graph. When the same pair co-appeared in multiple discussion groups, the corresponding edge weight was incremented accordingly. This procedure provides a transparent and reproducible method for transforming discussion records into editor co-participation networks.

### Question 2: Why is this network representation appropriate?

The weighted undirected representation preserves the strongest information available in the source data: whether two editors shared a discussion context, and how often this occurred. It is therefore well suited to the later tasks, because it preserves local clustering patterns created by repeated group discussion, provides a meaningful notion of tie strength, and supports the analysis of connectedness, small-world structure, and possible behavioural propagation.

### Question 3: What do the constructed networks show?

The three resulting networks differ substantially in scale. As shown in Table 1, `BOT_REQUESTS` contains 552 users and 2,424 edges, `PROPERTIES` contains 1,942 users and 9,198 edges, and `PROJECT_CHAT` contains 4,136 users and 40,255 edges. Despite these differences, all three contain a dominant giant component: the largest connected component includes 94.0% of nodes in `BOT_REQUESTS`, 83.3% in `PROPERTIES`, and 94.9% in `PROJECT_CHAT`. Table 1 also shows that `BOT_REQUESTS` has the highest density, whereas `PROJECT_CHAT` has the highest average degree.

This indicates that editor interaction in each dataset is concentrated within one main

conversational structure rather than being split into many isolated fragments. At the same time, **PROPERTIES** is more fragmented than the other two datasets, while **PROJECT\_CHAT** appears the most integrated. These networks therefore provide a suitable basis for the comparative analyses in Tasks B and C.

Table 1 – Core network properties of the three constructed editor networks

| Metric                  | BOT_REQUESTS | PROPERTIES | PROJECT_CHAT |
|-------------------------|--------------|------------|--------------|
| Users                   | 552          | 1942       | 4136         |
| Edges                   | 2424         | 9198       | 40255        |
| Largest component ratio | 0.940        | 0.833      | 0.949        |
| Density                 | 0.0159       | 0.0049     | 0.0047       |
| Average degree          | 8.783        | 9.473      | 19.466       |

## Part 1 - Task B: Network Metrics

**Question 1: What are the characteristic properties and relevant metrics of the three Wikidata editor networks?**

To characterise the three editor networks, I used a set of structural measures appropriate for locating them on the regular–small-world–random continuum: density, average degree, average clustering coefficient, average shortest path length, diameter, and degree heterogeneity. Since the networks are not fully connected, shortest-path-based measures were computed on the largest connected component, while global structural measures were calculated on the full graphs. This provides a consistent basis for comparing the observed networks with the expected properties of regular, small-world, and random structures.

As reported in Table 2, the networks are globally sparse but locally organised. Densities are low, at 0.0159 for **BOT\_REQUESTS**, 0.0049 for **PROPERTIES**, and 0.0047 for **PROJECT\_CHAT**, but average degree remains non-trivial, especially in **PROJECT\_CHAT**, where it reaches 19.47. Degree heterogeneity is also strong in all three networks: most editors have relatively low degree, while a small number of users occupy highly connected hub positions. This is reflected in the large degree standard deviations and maximum degrees, which rise from 226 in **BOT\_REQUESTS** to 627 in **PROPERTIES** and 1,166 in **PROJECT\_CHAT**. These patterns show that the networks are clearly not regular.

Local clustering is high in all three cases. The average clustering coefficients are 0.6576 for **BOT\_REQUESTS**, 0.4705 for **PROPERTIES**, and 0.6671 for **PROJECT\_CHAT**, while average shortest path lengths within the largest connected component remain low at 2.60, 2.90, and 2.74 respectively. This means that editors tend to form tightly overlapping local discussion groups, but those groups are still connected by relatively short routes across the wider network.

Table 2 – Structural metrics and random-graph baselines for the three editor networks

| Metric                   | BOT_REQUESTS | PROPERTIES | PROJECT_CHAT |
|--------------------------|--------------|------------|--------------|
| Density                  | 0.0159       | 0.0049     | 0.0047       |
| Avg. degree              | 8.7826       | 9.4727     | 19.4657      |
| Avg. clustering          | 0.6576       | 0.4705     | 0.6671       |
| Avg. shortest path (LCC) | 2.6028       | 2.9009     | 2.7406       |
| Diameter (LCC)           | 6            | 7          | 6            |
| Degree std               | 18.6147      | 24.5827    | 58.8715      |
| Max degree               | 226          | 627        | 1166         |
| $C_{\text{rand}}$        | 0.0157       | 0.0049     | 0.0047       |
| $L_{\text{rand}}$        | 3.1416       | 3.6185     | 3.0826       |
| Small-world index        | 50.5532      | 118.9937   | 160.2541     |

### Question 2: How different are these networks from random networks?

To assess the extent to which the observed networks depart from random structure, each empirical graph was compared with a matched Erdős–Rényi random graph  $G(n, m)$  containing the same number of nodes and edges. The comparison focused on the clustering coefficient and the average shortest path length, as these two quantities form the core diagnostic criteria in the conventional interpretation of small-world organisation.

The pattern is consistent across all three datasets. As shown in Figure 1 and summarised numerically in Table 2, in every case the real network has a clustering coefficient far above its random baseline, while average shortest path length remains of the same order of magnitude. For example, in BOT\_REQUESTS, clustering increases from 0.0157 in the random graph to 0.6576 in the real network, while average shortest path length changes only from 3.14 to 2.60. The same contrast appears in PROPERTIES and PROJECT\_CHAT. This is the defining empirical signature of small-world organisation: much stronger local clustering than random graphs, without a proportionate loss of global reach. The reported small-world indices are also all substantially greater than 1, reinforcing the same conclusion.

### Question 3: What do these results suggest about editor communication and social activity?

Overall, the results suggest that Wikidata editor communication is neither random nor uniformly structured. Instead, it combines strong local conversational cohesion with relatively efficient network-wide reach. Editors who participate in the same discussion contexts tend to form clustered neighbourhoods, which is consistent with repeated interaction around particular topics, pages, or subcommunities. At the same time, the low path lengths imply that these local clusters are not isolated: information, norms, or controversial discussions can potentially travel across the broader network in only a few steps.

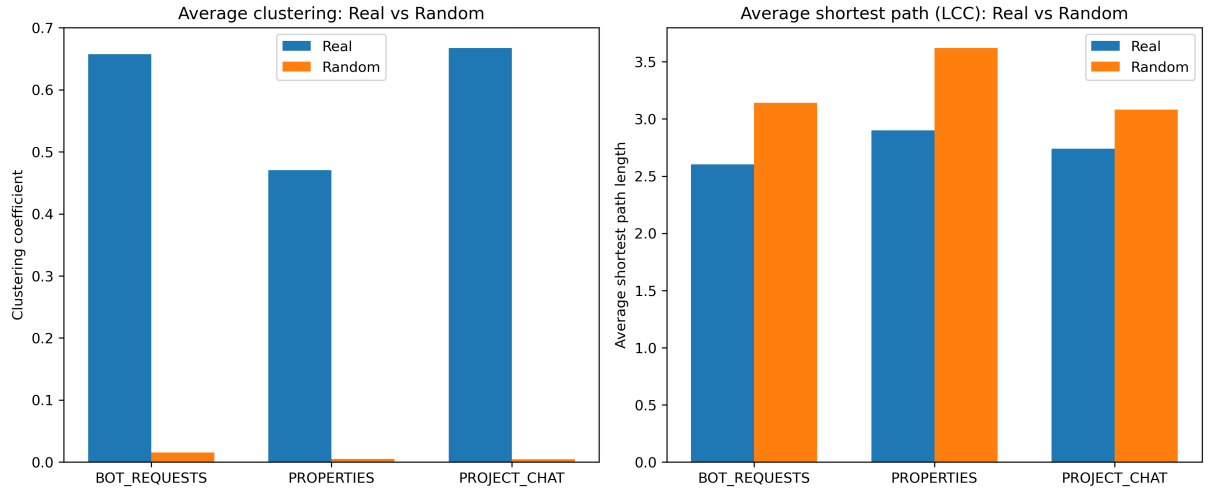


Figure 1 – Real vs random comparisons for the three editor networks.

There are also meaningful differences between the three datasets. **PROJECT\_CHAT** appears the most integrated and socially overlapping, with the highest average degree and the strongest hub structure. **BOT\_REQUESTS** is smaller but still highly clustered, indicating a compact conversational structure. **PROPERTIES** is more fragmented, with a smaller largest-component ratio, suggesting a more segmented interaction pattern. Overall, the three networks are best interpreted as small-world editor networks: they are unlike regular networks because of strong degree inequality and hub concentration, and unlike random networks because clustering remains far above random expectation.

## Part 1 - Task C: Epidemic Models

Question 1: How can the network be used to judge whether suspicious behaviour has plausibly not propagated yet?

This problem may be formulated as a network-based exposure assessment rather than a direct attempt to identify troll editors. The aim is to evaluate how structurally exposed surrounding editors are when one or two users exhibit unusually high activity on the same day. On this basis, non-propagation becomes less plausible when a suspicious editor is situated within a dense neighbourhood of nearby, similar, or repeatedly co-participating users. The monitoring framework therefore combines shortest-path proximity, neighbourhood similarity, shared neighbours, tie strength, and node activity, so that editors closer to the suspicious source and more embedded in the same local structure are treated as more plausible recipients of behavioural spread.

Task C: suspected editors and high-risk neighbors

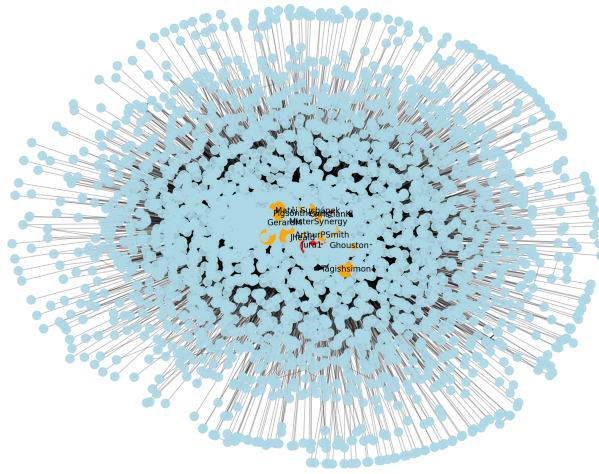


Figure 2 – High-risk neighbours around suspicious editors in `PROJECT_CHAT`, concentrated in the network core.

**Question 2: How should the Foundation generate a priority list of editors to inspect first?**

The same framework can be used to rank editors for follow-up inspection. Each editor receives a composite risk score based on proximity, similarity, shared neighbours, tie strength, and activity. The output is therefore not a binary accusation, but a priority list of structurally exposed editors. Figure 2 illustrates this using `PROJECT_CHAT`: suspicious editors are shown in red and high-risk neighbours in orange, concentrated in the dense network core. Table 3 shows that single-source non-propagation plausibility is Low in all three datasets, while dual-source plausibility remains Low in `PROPERTIES` and `PROJECT_CHAT` but is High in `BOT_REQUESTS`. Mean top-risk values are also higher in the medium and large networks.

Table 3 – Summary outputs for the Task C monitoring framework.

| Metric                              | <code>BOT_REQUESTS</code> | <code>PROPERTIES</code> | <code>PROJECT_CHAT</code> |
|-------------------------------------|---------------------------|-------------------------|---------------------------|
| Suspected user 1                    | Pasleim                   | Jura1                   | Jura1                     |
| Suspected user 2                    | Jura1                     | Infovarius              | ChristianKl               |
| Single non-propagation plausibility | Low                       | Low                     | Low                       |
| Single mean top risk                | 0.501526                  | 0.781998                | 0.686961                  |
| Dual non-propagation plausibility   | High                      | Low                     | Low                       |
| Dual mean top risk                  | 0.373678                  | 0.770508                | 0.646575                  |

Question 3: What does the epidemic-style simulation add, and what do the results imply across the three datasets?

A simple SIR-style simulation was used as an illustrative rather than predictive model of behavioural spread. Across the three datasets, interpretation depends on topology: **BOT\_REQUESTS** has a smaller and more traceable suspicious neighbourhood, **PROPERTIES** is more fragmented, and **PROJECT\_CHAT** has the strongest conditions for broader spillover because of its dense core and high clustering. Task C therefore complements Task B: proximity- and similarity-based monitoring is most important where the network structure makes wider spread more plausible.

## Comparative discussion across the three Wikidata editor networks

In summary, the three Wikidata editor networks are best characterised as occupying the small-world region of the regular–small-world–random continuum. Although all three networks are globally sparse, they combine relatively high clustering with short path lengths, and their degree distributions are strongly right-skewed, indicating hub-dominated rather than regular structures. **PROJECT\_CHAT** emerges as the most integrated network, **BOT\_REQUESTS** is smaller but comparatively compact, and **PROPERTIES** appears more fragmented. These structural contrasts also help to contextualise the findings of Task C: suspicious activity appears easier to trace locally in **BOT\_REQUESTS**, more partially contained in **PROPERTIES**, and more likely to diffuse across the network in **PROJECT\_CHAT**. Since ties are modelled as weighted, undirected co-participation within the same page-thread group, these results should be interpreted as characteristics of co-participation structure rather than as exact representations of temporal communication flows.

## Part2 - Task A: Spatial network construction and planarity

Question 1: What study area was selected, and why is it appropriate?

An approximately 1 km<sup>2</sup> study area in central Leeds was selected, and its driving road network was constructed for analysis. To identify a suitable area, Leeds accident data for 2012–2016 were combined, reduced to unique accident events, and scanned across a broader central-Leeds region using a 1 km × 1 km window. The final bounding box was [429900, 433200, 430900, 434200] in British National Grid coordinates, equivalent to [-1.547585, 53.794257, -1.532305, 53.803187] in longitude and latitude. The selected area contains 452 unique accidents, exceeding the coursework guideline, and Figure 3b shows that it lies within a compact central urban zone.

## Question 2: What are the main characteristics of the selected road network?

The selected road network was derived from OpenStreetMap using OSMnx, restricted to drivable roads, and projected into the British National Grid coordinate system. The resulting graph contains 211 nodes, 359 edges, 179 intersections, and a total edge length of 21.834 km. As shown in Table 4, the network exhibits a spatial diameter of 2930.01 m, an average street length of 60.82 m, a node density of 211.0 per km<sup>2</sup>, an intersection density of 179.0 per km<sup>2</sup>, and an edge density of 21,833.71 m per km<sup>2</sup>. Collectively, these measures characterise the study area as a dense, fine-grained urban road network with short links and frequent junctions, a pattern that is also evident in Figure 3a.

Table 4 – Summary statistics of the selected driving road network in the 1 km<sup>2</sup> study area.

| Metric                                      | Value    |
|---------------------------------------------|----------|
| Study area (km <sup>2</sup> )               | 1.0      |
| Total nodes                                 | 211      |
| Total edges                                 | 359      |
| Total intersections                         | 179      |
| Total edge length (km)                      | 21.834   |
| Spatial diameter (m)                        | 2930.01  |
| Average street length (m)                   | 60.82    |
| Node density (per km <sup>2</sup> )         | 211.0    |
| Intersection density (per km <sup>2</sup> ) | 179.0    |
| Edge density (m per km <sup>2</sup> )       | 21833.71 |
| Average circuitry                           | 1.6071   |
| Topologically planar                        | No       |

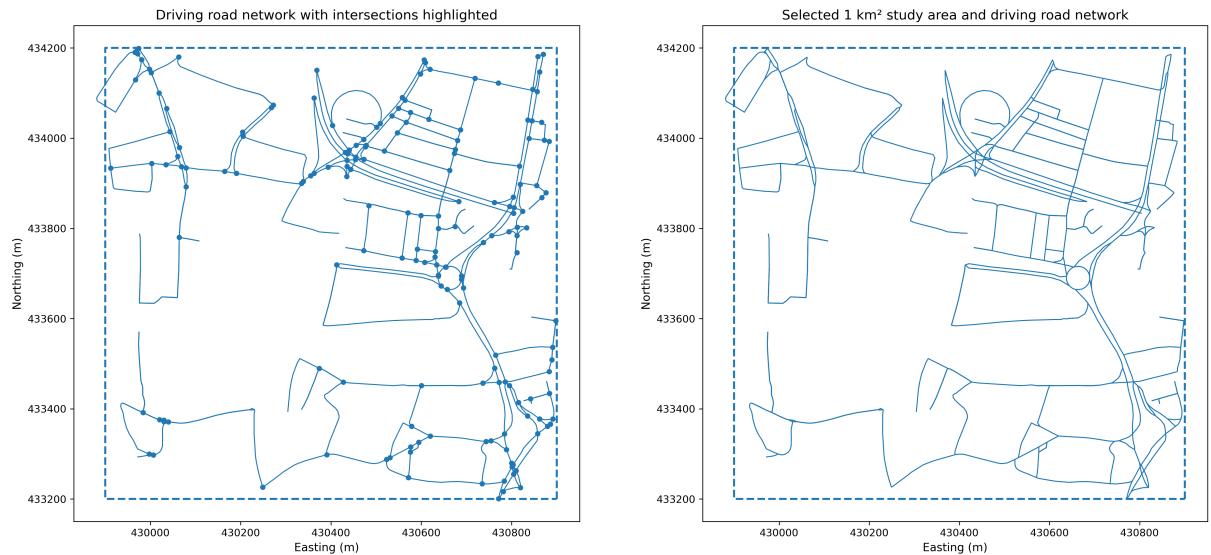
## Question 3: What is the average circuitry of the network, and what does it imply about movement efficiency?

The average circuitry of the network is 1.6071. Since circuitry measures the ratio between network travel distance and straight-line distance, a value above 1 indicates that movement through the road network involves detours relative to Euclidean travel. In this case, the value suggests that the network supports reasonably efficient local movement, but not perfectly direct travel, which is consistent with the compact urban street layout shown in Figure 3a.

## Question 4: Is the road network planar, and why does this matter?

A graph-theoretic planarity test (Brandes, 2009) returned No for the undirected road network. This should not be read as a flaw in the data, but as a realistic feature of urban transport systems. A road network is strictly planar only if edges intersect exclusively at shared nodes and no roads cross in map view without forming an actual

junction. In practice, bridges, underpasses, slip roads, and grade-separated junctions often violate this condition. The selected Leeds road network should therefore be described as not strictly planar, both in graph-theoretic and practical geographic terms. Figures 3 helps illustrate this distinction by showing that not every apparent crossing in the mapped layout corresponds to a true planar intersection.



(a) Driving network with intersections highlighted.

(b) Selected 1 km<sup>2</sup> study area and extracted driving network.

Figure 3 – Road network visualisations of the selected Leeds study area.

## Part2 - Task B: Road accidents on the network

### Question 1: How are road accidents distributed on the selected road network?

Task B investigates how the 452 unique road accidents recorded within the selected 1 km<sup>2</sup> central Leeds study area are distributed over the driving road network during 2012–2016. To analyse this, accidents were snapped to the road network and then aggregated by network arc. Figure 4 shows the resulting accident counts by road edge. The pattern is highly uneven: most arcs have very low counts, while a small subset of arcs carry noticeably higher totals. The highest-count arcs are not randomly scattered, but occur within connected parts of the network, especially near central junction environments and along a few important connectors. This indicates that accident risk is concentrated on particular corridors and junction approaches rather than being spread evenly across all streets.

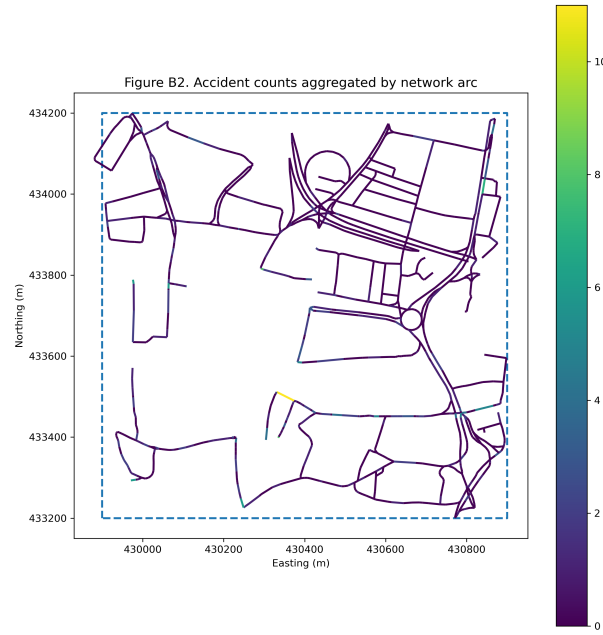


Figure 4 – Accident counts aggregated by road edge, showing uneven concentration across the network.

### Question 2: Do roads with many accidents tend to be connected to other high-accident roads?

To assess whether roads with high accident counts tend to be connected to other high-accident roads, Moran's I (Cliff & Ord, 1981) and the global network K-function (Ó'Sullivan & Unwin, 2010) were used. Moran's I was 0.1437, compared with an expected value of -0.0007 under randomness, with  $p = 0.001$  and  $z = 5.9278$ , indicating a positive and statistically significant autocorrelation in accident counts across connected roads. As shown in Figure 5c, the observed K-function also remains above the simulated upper envelope over much of the distance range, particularly beyond about 1.3 km, indicating that accidents are more clustered along the network than would be expected under a random network process. Taken together, these results suggest that accident risk is not confined to isolated road segments, but is spatially structured across connected parts of the local road network.

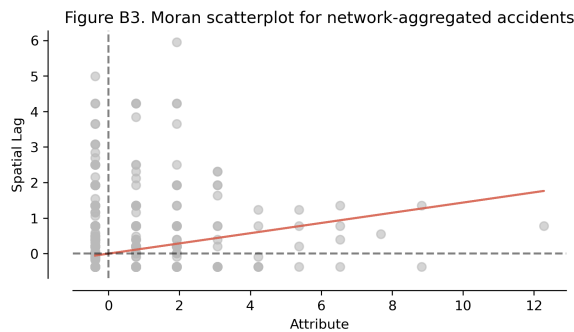
### Question 3: Do accidents occur nearer to intersections or further along roads?

To answer the third question, each snapped accident was expressed as the fraction of arc length from the nearest intersection. A value close to 0 indicates an accident very near a junction, whereas values closer to 0.5 indicate a more mid-segment location. The histogram in Figure 5d is clearly concentrated toward smaller values, indicating that accidents occur disproportionately close to intersections. The summary statistics in Table 5 support the same conclusion: the mean fraction is 0.2212, the median is 0.1723, 34.96% of

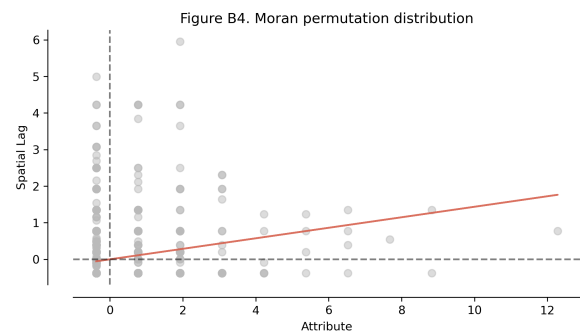
accidents occur within the nearest 10% of arc length from an intersection, and 51.99% occur within the nearest 20%. By contrast, only 21.24% occur beyond 40% of arc length. This indicates that accidents are substantially more likely to occur near intersections than along mid-block sections of road, which is plausible in an urban driving network where junctions concentrate turning, merging, and crossing conflicts.

Table 5 – Key summary outputs for the accident-network analysis

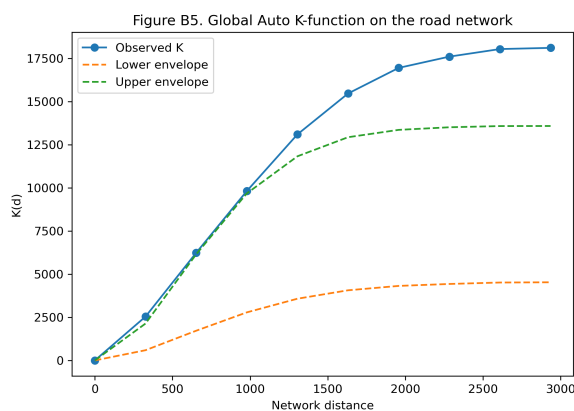
| Measure                                   | Value   |
|-------------------------------------------|---------|
| Unique accidents (2012–2016)              | 452     |
| Moran’s $I$                               | 0.1437  |
| Expected Moran’s $I$                      | -0.0007 |
| Moran permutation $p$ -value              | 0.001   |
| Moran $z$ -score                          | 5.9278  |
| Mean fraction from nearest intersection   | 0.2212  |
| Median fraction from nearest intersection | 0.1723  |
| % within nearest 10% of arc length        | 34.96   |
| % within nearest 20% of arc length        | 51.99   |
| % beyond 40% of arc length                | 21.24   |



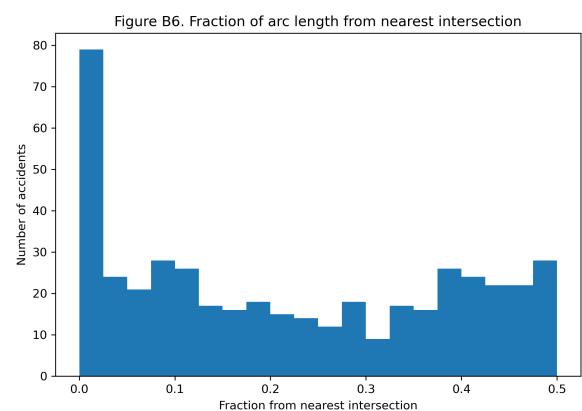
(a) Moran scatterplot for network-aggregated accident counts.



(b) Permutation distribution for Moran’s  $I$ .



(c) Global network K-function with simulation envelope.



(d) Fraction of arc length from the nearest intersection for snapped accident events.

Figure 5 – Accident clustering and intersection-proximity diagnostics.

## Part2 - Task C: Voronoi diagrams

### Question 1: How were the initial four seed points selected?

The four initial seed points were selected using a risk-aware and spatially distributed strategy. Candidate nodes were prioritised if they were relatively far from accident-prone roads identified in Task B, reasonably well connected within the network, and sufficiently separated from one another in space. This was intended to create four starting locations that were both accessible and less exposed to the most accident-concentrated parts of the road system. As shown in Figure 6a, the selected seeds are distributed across the study area rather than concentrated in a single part of the network

### Question 2: What kind of Voronoi diagram is most useful here, and what does it show?

The most appropriate partitioning method in this context is a node-based network Voronoi diagram, in which each network node is assigned to the nearest seed according to shortest-path distance along the road network. This is more useful than a planar Voronoi diagram because marathon accessibility should be understood in terms of movement along roads rather than straight-line Euclidean distance. As shown in Figure 6b, the resulting four cells are spatially distinct and provide a reasonable accessibility-based partition of the selected network. However, they also differ substantially in their internal network extent, which is an important constraint for route design.

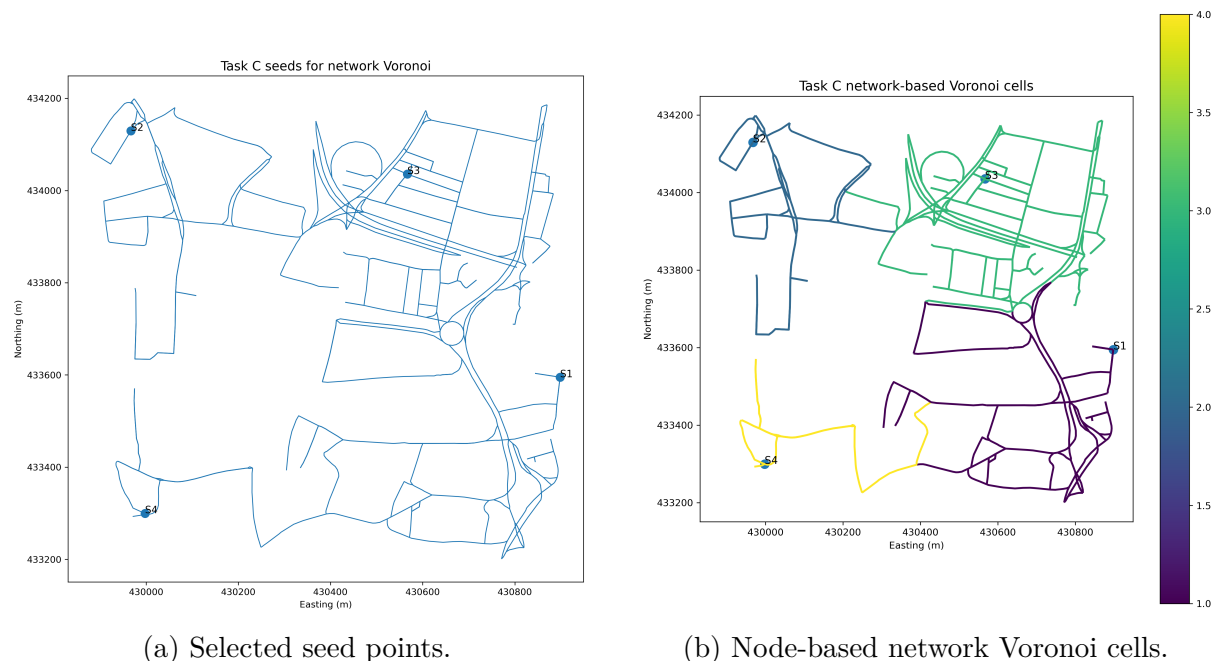


Figure 6 – Seed selection and network Voronoi partition.

Table 6 – Initial route-feasibility summary and redesign comparison for the candidate cells

| Cell | Nodes | Edges | Road<br>length (km) | Connected | Components | Tested | Success<br>within<br>tolerance | Redesign<br>success |
|------|-------|-------|---------------------|-----------|------------|--------|--------------------------------|---------------------|
| 1    | 62    | 83    | 5.208               | 1         | 1          | 1      | 0                              | 0                   |
| 2    | 32    | 43    | 3.673               | 1         | 1          | 1      | 0                              | 0                   |
| 3    | 103   | 142   | 10.081              | 1         | 1          | 1      | 0                              | 0                   |
| 4    | 14    | 16    | 1.582               | 1         | 1          | 1      | 0                              | 0                   |

**Question 3: Can 2 or 3 cells support a closed route of approximately 42 km, and what do the route-search results show?**

Following the coursework specification, the route-search procedure was first applied to a subset of cells as an initial feasibility test. No feasible closed route of approximately 42km was found in any of the tested cells. This result is best explained by the limited road extent available within each partition rather than by the route-search procedure itself. Even the larger cells contain only a small amount of drivable road relative to the marathon target distance. The visual route-search outputs for the initially tested cells are provided in the appendix, where they show that these sub-networks are compact and highly local in scale rather than suitable for marathon-length closed routes.

**Question 4: Can the route search be extended to all four cells?**

The route search was then extended to all four cells. As summarised in Table 6, none of the four cells produced a feasible closed route close to 42 km. The total road lengths available within the initial cells are only 5.208 km, 3.673 km, 10.081 km, and 1.582 km respectively, and all cells record `success_within_tolerance = 0`. This indicates that the problem is structural rather than computational: once the selected 1 km<sup>2</sup> study area is divided into four cells, each resulting sub-network becomes too small to support a realistic marathon-length closed route. The full four-cell route-search results are shown in Figure 7.

**Question 5: If some cells have no such route, what modification should be considered, and what does the repeated analysis show?**

Because the initial four-cell design was infeasible, the analysis was repeated using a redesigned seed configuration intended to create a more balanced partition. However, as shown in Table 6, the redesign did not improve route feasibility. No feasible 42 km closed route was found either before or after redesign. This suggests that the main constraint is structural rather than methodological. In particular, changing the seed locations within the same compact 1 km<sup>2</sup> study area cannot overcome the limited total amount of drivable

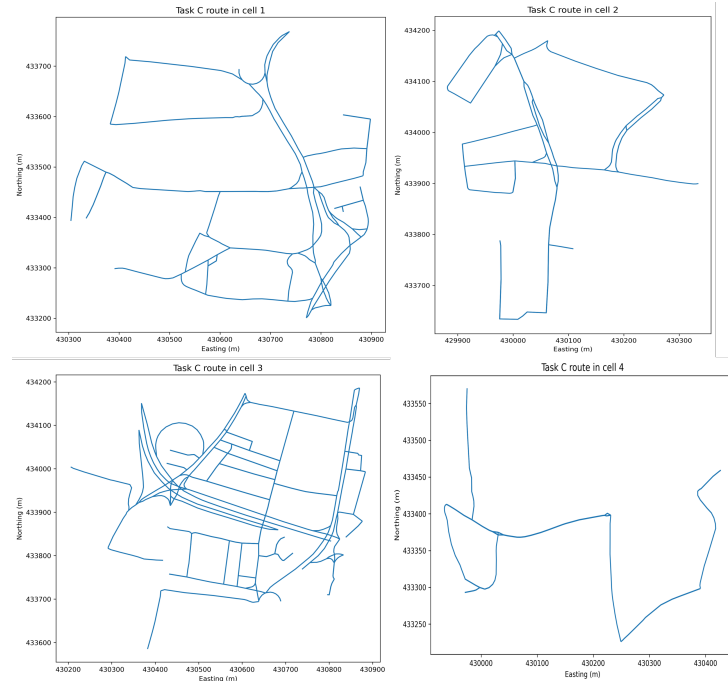


Figure 7 – Route-search results for the four initial Voronoi cells. All four sub-networks are too limited in spatial extent to support a realistic closed route of approximately 42 km.

road available in each cell. A more realistic improvement would therefore involve either enlarging the study area substantially or reducing the number of marathon cells.

## Comparative discussion across the Leeds road-network tasks

Overall, the three Leeds road-network tasks indicate that the selected  $1 \text{ km}^2$  central study area is well suited to local transport-network analysis, but not to large-scale marathon route design. Task A showed that the area contains a dense and intersection-rich driving network with short links, frequent junctions, an average circuitry of 1.6071, and a total road length of 21.834 km. Task B further demonstrated that accident occurrence is not randomly distributed, but instead concentrated on a limited subset of roads, positively autocorrelated across connected arcs, more clustered than expected under the network K-function, and disproportionately associated with intersections. Task C, by contrast, revealed a clear mismatch between the spatial scale of the selected network and the marathon-planning objective. Although the node-based network Voronoi diagram is the most appropriate partitioning approach in this context, because accessibility should be evaluated along roads rather than by straight-line distance, dividing such a compact network into four cells leaves each cell too limited in extent to support a realistic 42 km closed route. The repeated analysis with redesigned seed points did not alter this conclusion. The main limitation therefore lies not in the Voronoi approach itself, but in the spatial scale of the selected study area.

# Appendix

## Part 1 – Task A (Extended Construction Details)

### Question 1: What was the algorithmic approach?

The network construction followed a simple and reproducible workflow:

1. Read the CSV file and retain the `page`, `thread`, and `username` fields.
2. Remove missing values and exact duplicate rows.
3. Group the data by (`page`, `thread`) so that each group represents one discussion context.
4. Extract the unique users from each group.
5. Generate all unordered user pairs within the group.
6. Add each pair to the graph, or increase its weight if the pair has already appeared elsewhere.
7. Build the final weighted undirected graph and compute summary statistics.

This separates preprocessing, pair generation, and graph construction, and makes the same workflow consistent across all three datasets.

### Question 2: What supplementary results support the validity of the networks?

The expanded summary statistics in Figure 8 confirm that the three datasets differ clearly in size but all produce coherent interaction networks. After cleaning, the datasets contain 2,981, 13,398, and 48,551 rows respectively, and the resulting networks contain 29, 304, and 203 connected components. The largest connected components contain 519, 1,618, and 3,924 users respectively. These values support the interpretation that each dataset contains one dominant conversational core rather than only disconnected fragments.

### Question 3: What do the network visualisations show?

The largest-connected-component visualisations provide a useful qualitative check on the construction results. As shown in Figure 8, all three networks display a clear central interaction core together with a less connected periphery. `BOT_REQUESTS` appears relatively sparse, `PROPERTIES` appears broader and more fragmented, and `PROJECT_CHAT` shows the most concentrated and densely connected core. These visual patterns are consistent with the summary statistics in Table 7 and support the validity of the construction.

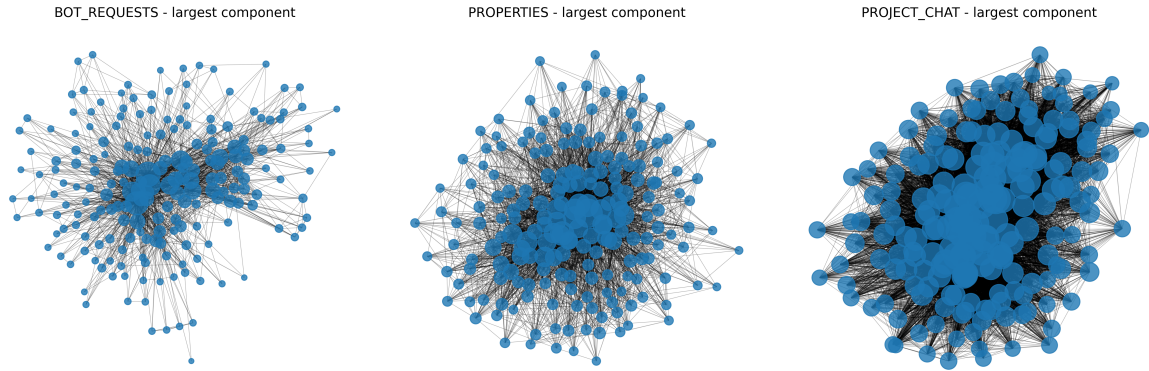


Figure 8 – Largest connected components of the editor networks. Node size is proportional to the degree or weighted degree of each node, and edges represent co-participation of two editors in the same page-thread discussion group.

Table 7 – Expanded summary statistics for the three constructed editor networks

| Metric                  | BOT_REQUESTS | PROPERTIES | PROJECT_CHAT |
|-------------------------|--------------|------------|--------------|
| Rows after cleaning     | 2981         | 13398      | 48551        |
| Users                   | 552          | 1942       | 4136         |
| Edges                   | 2424         | 9198       | 40255        |
| Pages                   | 108          | 2544       | 131          |
| Threads                 | 1113         | 5785       | 14916        |
| Page-thread groups      | 1126         | 6681       | 15412        |
| Components              | 29           | 304        | 203          |
| Isolates                | 25           | 285        | 195          |
| Largest component size  | 519          | 1618       | 3924         |
| Largest component ratio | 0.940        | 0.833      | 0.949        |

## Part 1 – Task B (supplementary metric details)

Question 1: What additional metric details support the characterisation of the networks?

The main analysis focused on density, average degree, clustering, shortest path length, diameter, and degree heterogeneity because these are sufficient to place the three editor networks on the regular–small-world–random spectrum. In addition, the analysis also examined maximum degree and degree distributions, both of which confirmed that the networks are strongly right-skewed rather than degree-homogeneous. As reported in Table 8, the largest connected component was used for shortest-path-based measures because none of the three graphs is fully connected. This avoids undefined path lengths while still focusing on the dominant interaction structure already established in Task A.

## Question 2: What supplementary evidence supports the small-world interpretation?

The clearest evidence is the contrast between very high clustering in the real networks and much lower clustering in matched random graphs. At the same time, average shortest path lengths remain low and broadly comparable to the random baselines. This combination holds for `BOT_REQUESTS`, `PROPERTIES`, and `PROJECT_CHAT`, which is why all three are interpreted as small-world rather than random networks. As summarised in Table 8, the reported small-world indices are all substantially greater than 1 and should be read as supporting evidence rather than the sole basis for classification.

## Question 3: What do the figures show?

As shown in Figure 9, the degree distribution histograms display a strongly right-skewed pattern in all three datasets: most editors have low degree, while a small number of users act as highly connected hubs. This pattern is especially pronounced in `PROJECT_CHAT`, where the upper tail extends much further than in the other two networks, consistent with its maximum degree of 1,166 and much larger degree standard deviation. `PROPERTIES` also shows substantial heterogeneity, while `BOT_REQUESTS` has a shorter tail but still departs clearly from any approximately uniform degree pattern.

These histograms reinforce two conclusions: the networks are not regular, and connectivity is concentrated in a minority of hub-like editors who help preserve short global path lengths. The comparison between the real networks and the matched random graphs also shows that clustering remains much higher in the observed data while path lengths stay relatively short. Visually, this is the key reason the networks are best placed in the small-world region of the continuum. Figure 9 therefore complements the numerical evidence in Table 8 by showing the inequality of node connectivity, while Table 8 summarises the clustering and path-length pattern that defines small-world structure.

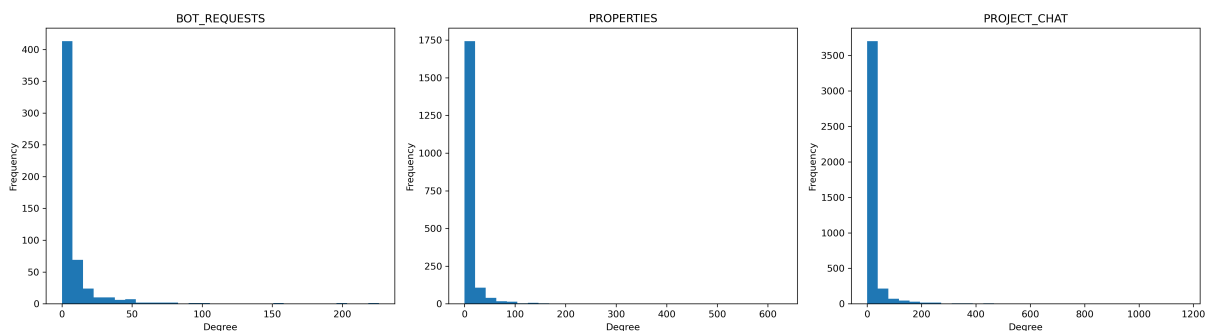


Figure 9 – All three degree distributions are strongly right-skewed, with most editors concentrated at low degree and only a small minority exhibiting very high degree. The longest upper tail appears in `PROJECT_CHAT`, which is consistent with the greatest degree heterogeneity among the three networks.

Table 8 – Core structural metrics and random-graph baselines

| Metric                   | BOT_REQUESTS | PROPERTIES | PROJECT_CHAT |
|--------------------------|--------------|------------|--------------|
| $n$                      | 552          | 1942       | 4136         |
| $m$                      | 2424         | 9198       | 40255        |
| Density                  | 0.0159       | 0.0049     | 0.0047       |
| Avg. degree              | 8.7826       | 9.4727     | 19.4657      |
| Largest component size   | 519          | 1618       | 3924         |
| Largest component ratio  | 0.9402       | 0.8332     | 0.9487       |
| Avg. clustering          | 0.6576       | 0.4705     | 0.6671       |
| Avg. shortest path (LCC) | 2.6028       | 2.9009     | 2.7406       |
| Diameter (LCC)           | 6            | 7          | 6            |
| Degree std               | 18.6147      | 24.5827    | 58.8715      |
| Max degree               | 226          | 627        | 1166         |
| $C_{\text{rand}}$        | 0.0157       | 0.0049     | 0.0047       |
| $L_{\text{rand}}$        | 3.1416       | 3.6185     | 3.0826       |
| Small-world index        | 50.5532      | 118.9937   | 160.2541     |

## Part 1 – Task C (supplementary monitoring details)

Question 1: What additional methodological details support the monitoring framework?

The Task C framework was designed as a triage tool rather than a classifier. Each editor was assigned a risk score based on a combination of shortest-path proximity, neighbourhood similarity, shared neighbours, repeated co-participation strength, and node activity. These components were chosen because they capture different mechanisms through which controversial behaviour could plausibly spread across a co-participation network: direct local exposure, similarity of discussion context, overlap of conversational circles, repeated interaction, and onward reach through highly connected positions. A high score does not imply guilt; rather, it means that an editor is more structurally exposed and should be checked earlier if monitoring resources are limited.

Question 2: What do the risk-network figures show across the three datasets?

The risk-network figures visualise suspicious editors in red and their highest-risk neighbouring editors in orange. In all three datasets, the highlighted high-risk nodes cluster around the suspicious editors rather than appearing randomly distributed. This is important because it shows that the ranking procedure is driven by local exposure structure rather than by arbitrary node selection.

In BOT\_REQUESTS (Figure 10), the suspicious editors and high-risk neighbours are concentrated within a relatively compact central neighbourhood. In PROPERTIES (Figure 11), the suspicious editors also sit near the graph’s core, but the surrounding

high-risk structure appears broader and less compact. In **PROJECT\_CHAT** (Figure 2 in the main text), the red and orange nodes are embedded in a very dense central core, suggesting that monitoring is especially important in this dataset because suspicious activity has more structurally plausible routes for wider spillover. The corresponding summary outputs are reported in Table 9.

Table 9 – Summary outputs for the monitoring framework

| Metric                              | BOT_REQUESTS | PROPERTIES | PROJECT_CHAT |
|-------------------------------------|--------------|------------|--------------|
| Suspected user 1                    | Pasleim      | Jura1      | Jura1        |
| Suspected user 2                    | Jura1        | Infovarius | ChristianKl  |
| Single non-propagation plausibility | Low          | Low        | Low          |
| Single mean top risk                | 0.501526     | 0.781998   | 0.686961     |
| Dual non-propagation plausibility   | High         | Low        | Low          |
| Dual mean top risk                  | 0.373678     | 0.770508   | 0.646575     |

### Question 3: What do the SIR figures show across the three datasets?

The SIR figures provide a short-run illustration of how suspicious behaviour could propagate if one or two editors act as initial sources. In all three datasets, the susceptible population remains the largest group over the observed time window, which indicates that spread is not immediate or explosive. However, the infected and recovered curves still rise over time, showing that propagation is plausible rather than absent.

In **BOT\_REQUESTS** (Figure 12), both infection and recovery remain low throughout the simulation, and the difference between the single-source and dual-source cases is relatively modest. In **PROPERTIES** (Figure 13), the infected curves rise more clearly, especially in the two-source case. In **PROJECT\_CHAT** (Figure 14), the same pattern is strongest: the susceptible population remains dominant, but infected and recovered counts increase steadily, and the dual-source case produces broader spread than the single-source case. These figures reinforce the interpretation that simultaneous suspicious activity by two users reduces confidence that the behaviour remains confined.

### Question 4: How do these results relate back to Task B?

The supplementary figures reinforce the interpretation from Task B rather than replacing it. Networks with strong clustering, short paths, hub concentration, and a large connected core are not only socially organised but also structurally capable of supporting broader behavioural exposure. This is why **PROJECT\_CHAT** shows the strongest visual evidence of wider spillover, while **BOT\_REQUESTS** remains the most locally interpretable and **PROPERTIES** occupies an intermediate position between compact spread and partial

confinement. Table 9 is consistent with this interpretation, since the dual-source non-propagation plausibility remains low in **PROPERTIES** and **PROJECT\_CHAT** but is higher in **BOT\_REQUESTS**.

Task C: suspected editors and high-risk neighbors

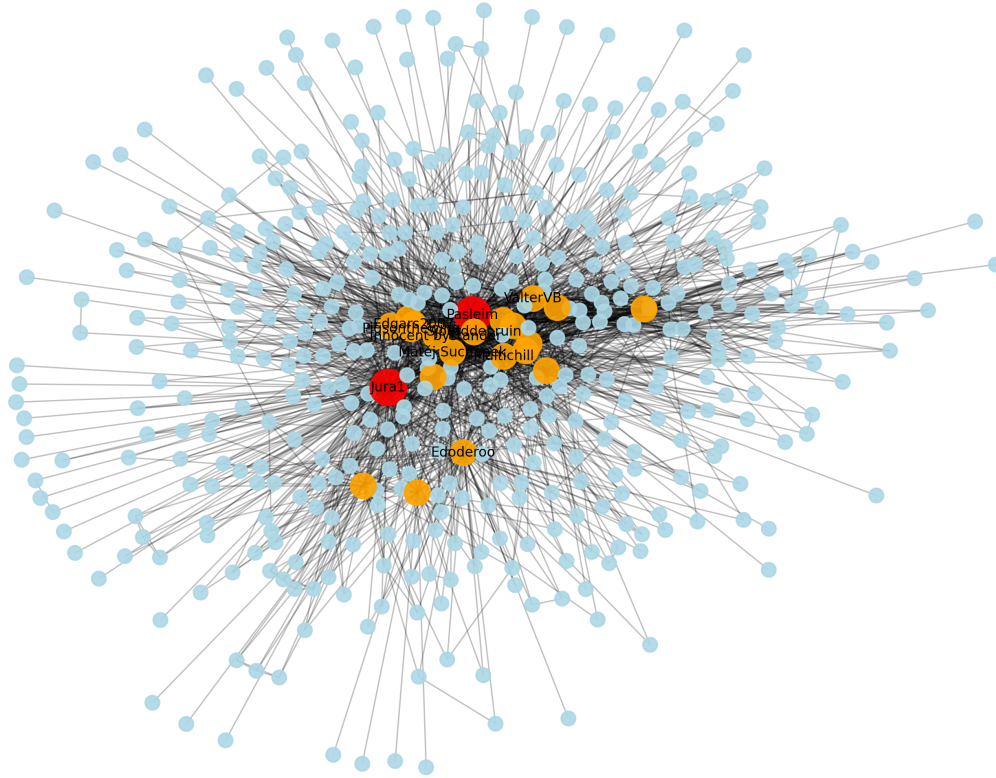


Figure 10 – High-risk neighbours around suspicious editors in **BOT\_REQUESTS**. The highlighted neighbourhood is relatively compact and visually easier to trace, consistent with a smaller and more interpretable local exposure structure.

## Part 2 – Task A (supplementary spatial-network details)

### Question 1: What additional details support the study-area selection?

The selected study area was derived by combining accident data from 2012–2016 and converting casualty-level records into unique accident events using the accident reference number. A sliding  $1\text{ km} \times 1\text{ km}$  window was then moved across a broader search region around central Leeds, and the number of unique accidents within each candidate window was counted. The final area was selected because it combined a central location with a sufficiently high accident total for meaningful downstream analysis. The final bounding box was  $[429900, 433200, 430900, 434200]$  in British National Grid coordinates, equivalent

## Task C: suspected editors and high-risk neighbors

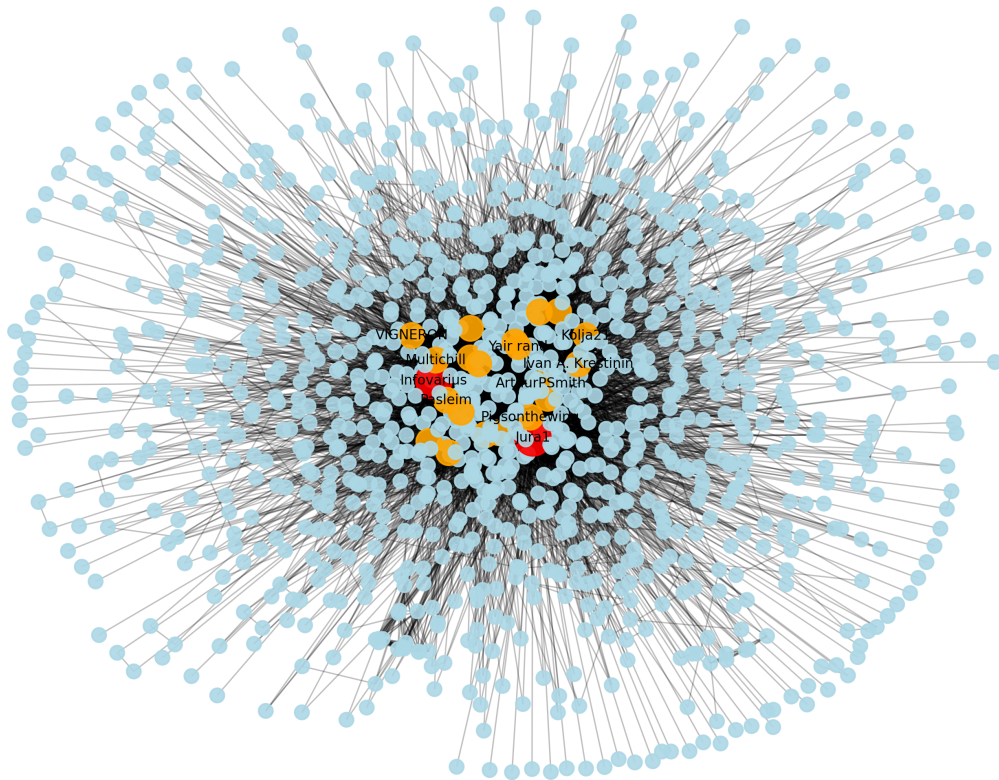


Figure 11 – High-risk neighbours around suspicious editors in **PROPERTIES**. High-risk nodes remain concentrated near the suspicious editors, but the surrounding structure is broader and somewhat less compact than in **BOT\_REQUESTS**.

to  $[-1.547585, 53.794257, -1.532305, 53.803187]$  in longitude and latitude, and the area contains 452 unique accidents over the five-year period. This shows that the selected  $1 \text{ km}^2$  area was not arbitrary, but chosen to satisfy both the spatial and event-density requirements of the coursework.

Figure 15 corresponds to the study-area selection visualisation. It shows the chosen central-Leeds window together with the extracted driving road network. As a supplementary figure, it is useful because it makes the selection procedure transparent: the reader can see both the compact geographic extent and the urban context of the chosen area.

**Question 2: What additional implementation details support the road-network construction?**

The road network was downloaded from OpenStreetMap with OSMnx and filtered to include only roads suitable for driving, excluding walking paths and private roads. The network was then projected to the British National Grid so that all length calculations could

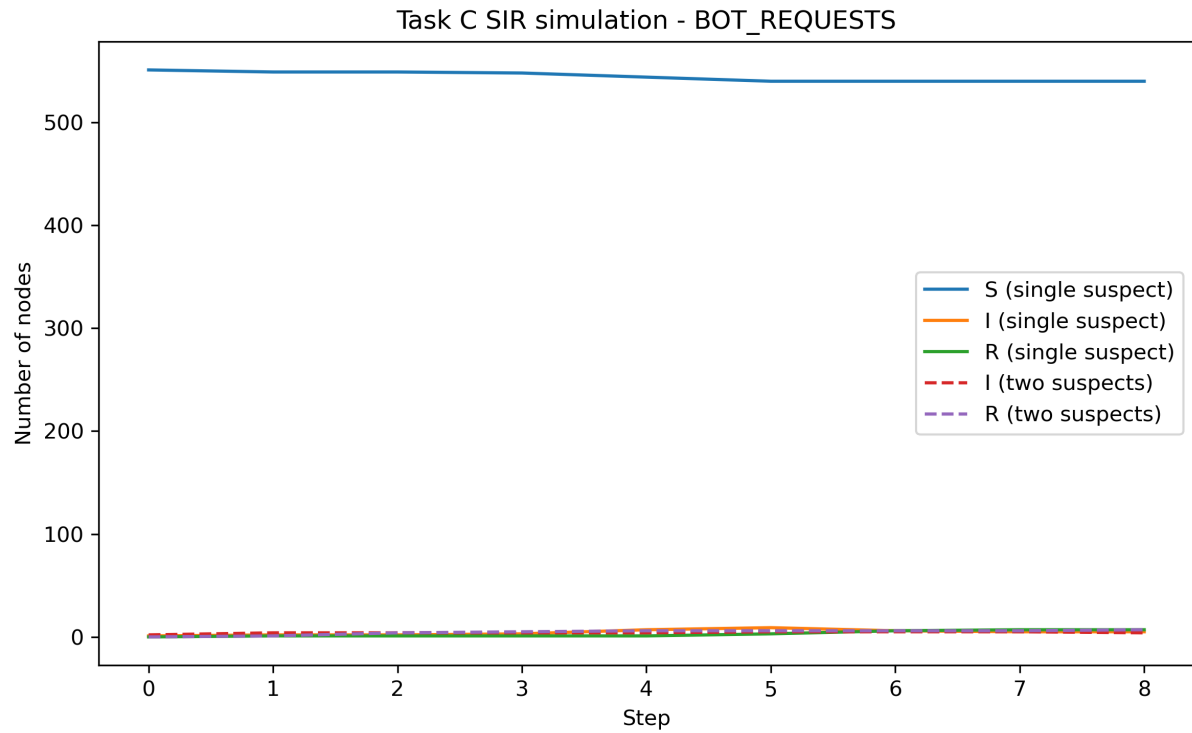


Figure 12 – SIR simulation for suspicious-editor propagation in **BOT\_REQUESTS**. Propagation remains limited over the short simulation window, and the difference between single-source and dual-source scenarios is comparatively modest.

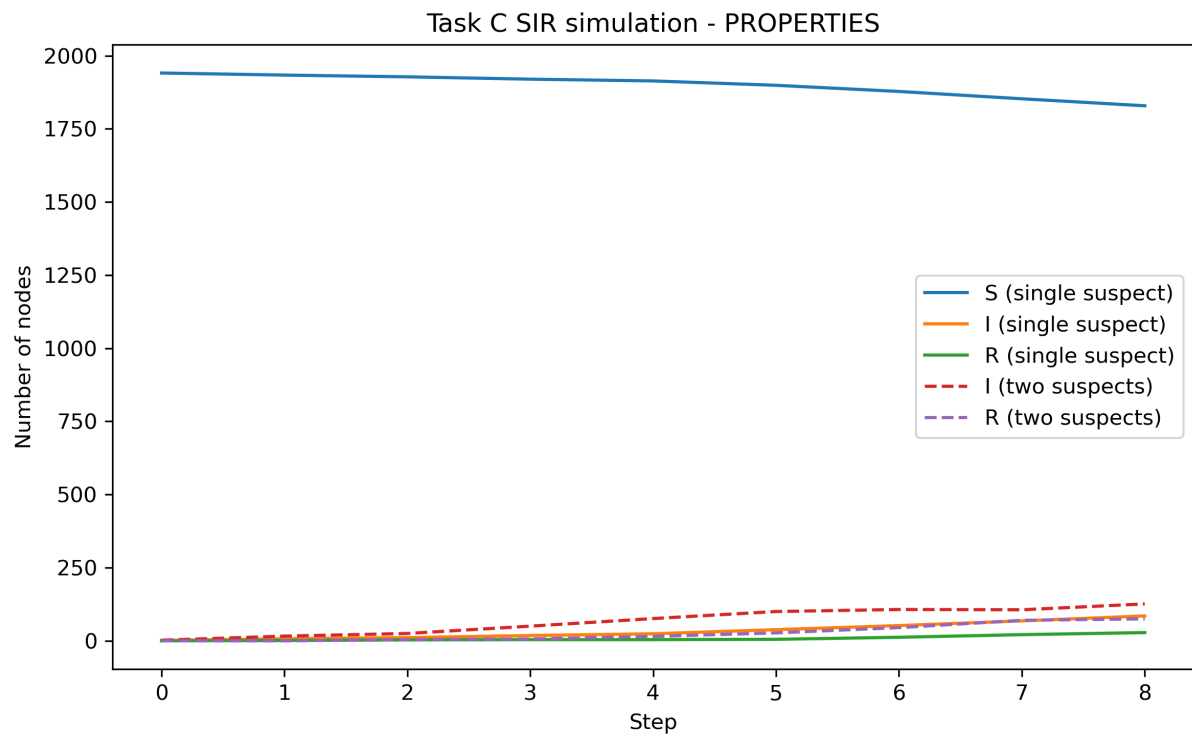


Figure 13 – SIR simulation for suspicious-editor propagation in **PROPERTIES**. The dual-source infection curve rises more clearly than the single-source curve, indicating broader simulated exposure when two suspicious editors are active simultaneously.

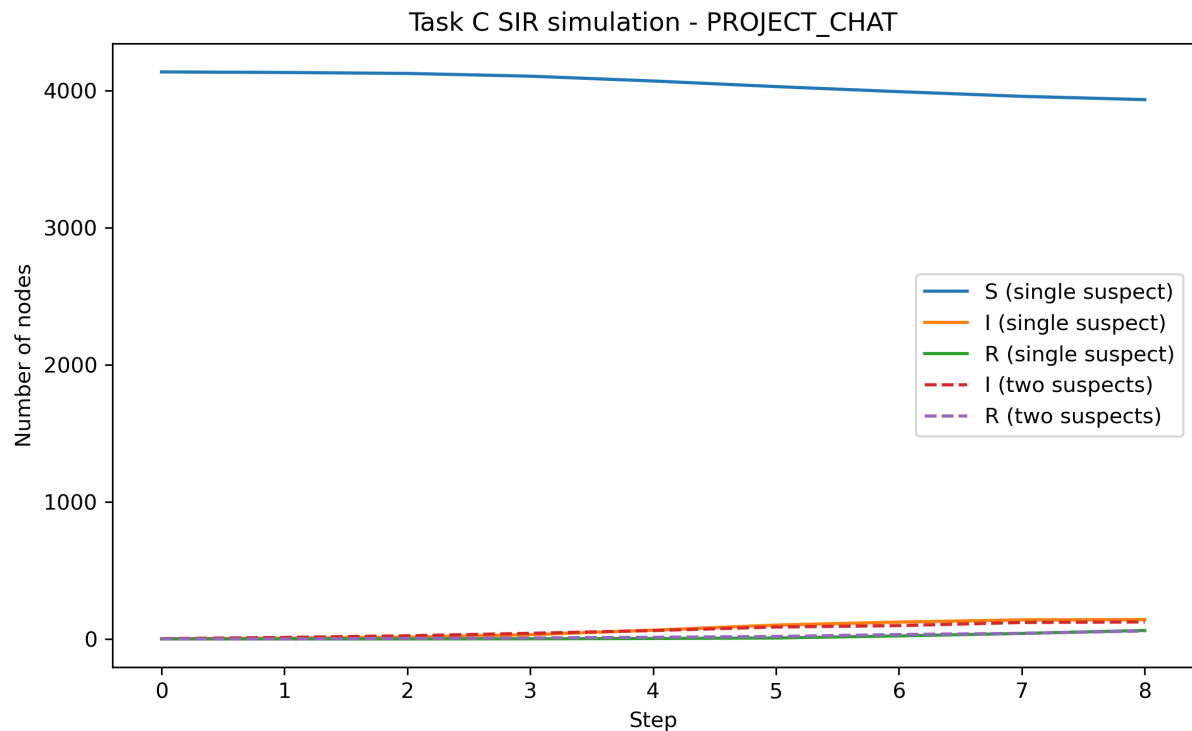


Figure 14 – SIR simulation for suspicious-editor propagation in `PROJECT_CHAT`. The susceptible population remains dominant over the short simulation window, but infected and recovered counts increase steadily. The dual-source scenario produces broader spread than the single-source case, suggesting that simultaneous suspicious activity by two editors lowers the plausibility that the behaviour remains confined.

be made in metres. The resulting graph contains 211 nodes, 359 edges, 179 intersections, and a total road length of 21.834km. These quantities are reported together in Table 10, which expands the main-text table by keeping the complete set of descriptive statistics used in the analysis.

Figure 16 shows the internal structure of the road network with intersections highlighted. Compared with the study-area overview, this panel is more useful for understanding the local articulation of the graph. It shows a dense and fine-grained urban network with many closely spaced intersections and short road segments.

### Question 3: What supplementary interpretation can be given for efficiency and planarity?

The average circuitry value of 1.6071 indicates that travel through the network requires detours relative to straight-line distance, but not excessive ones. In an urban setting, this is expected: movement is constrained by street geometry, turning structure, and block layout. The value therefore supports the interpretation that the selected network is reasonably efficient for local movement while still reflecting the indirectness of real roads.

The graph-theoretic planarity check classified the network as non-planar. This should not be read as a flaw in the data or method, but as a realistic feature of urban road systems. Apparent crossings in map space may correspond to bridges, underpasses, or grade-separated junctions rather than true intersections. Figures 15 and 16 help illustrate this distinction by showing that the visible road layout and the highlighted intersections do not imply that every spatial crossing should be interpreted as a planar node. The non-planarity result is therefore methodologically meaningful rather than problematic.

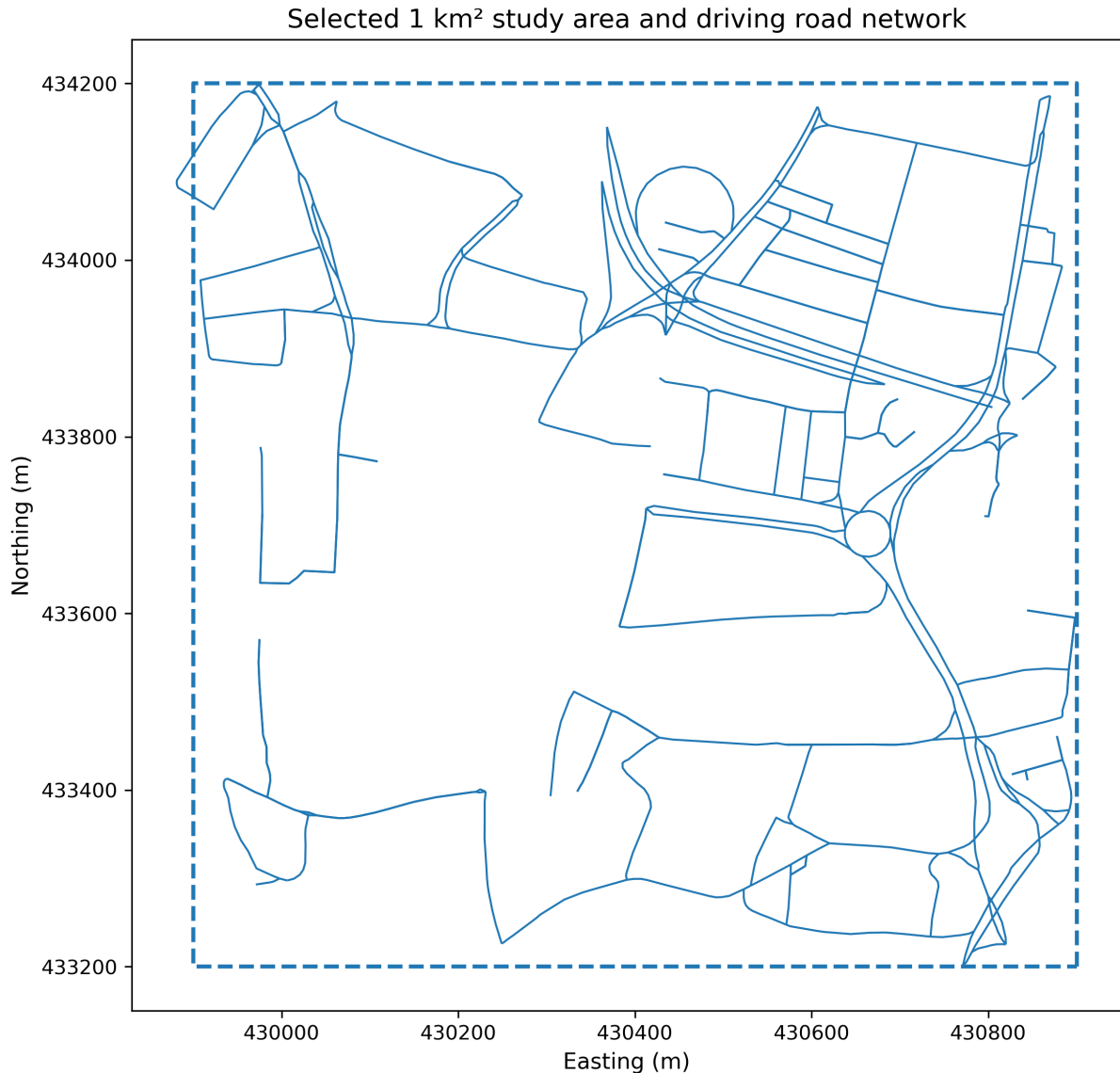


Figure 15 – Selected 1 km<sup>2</sup> study area in central Leeds and extracted driving road network. This figure shows the location and extent of the chosen central Leeds study area and confirms that the selected road network covers a compact, accident-rich urban zone.

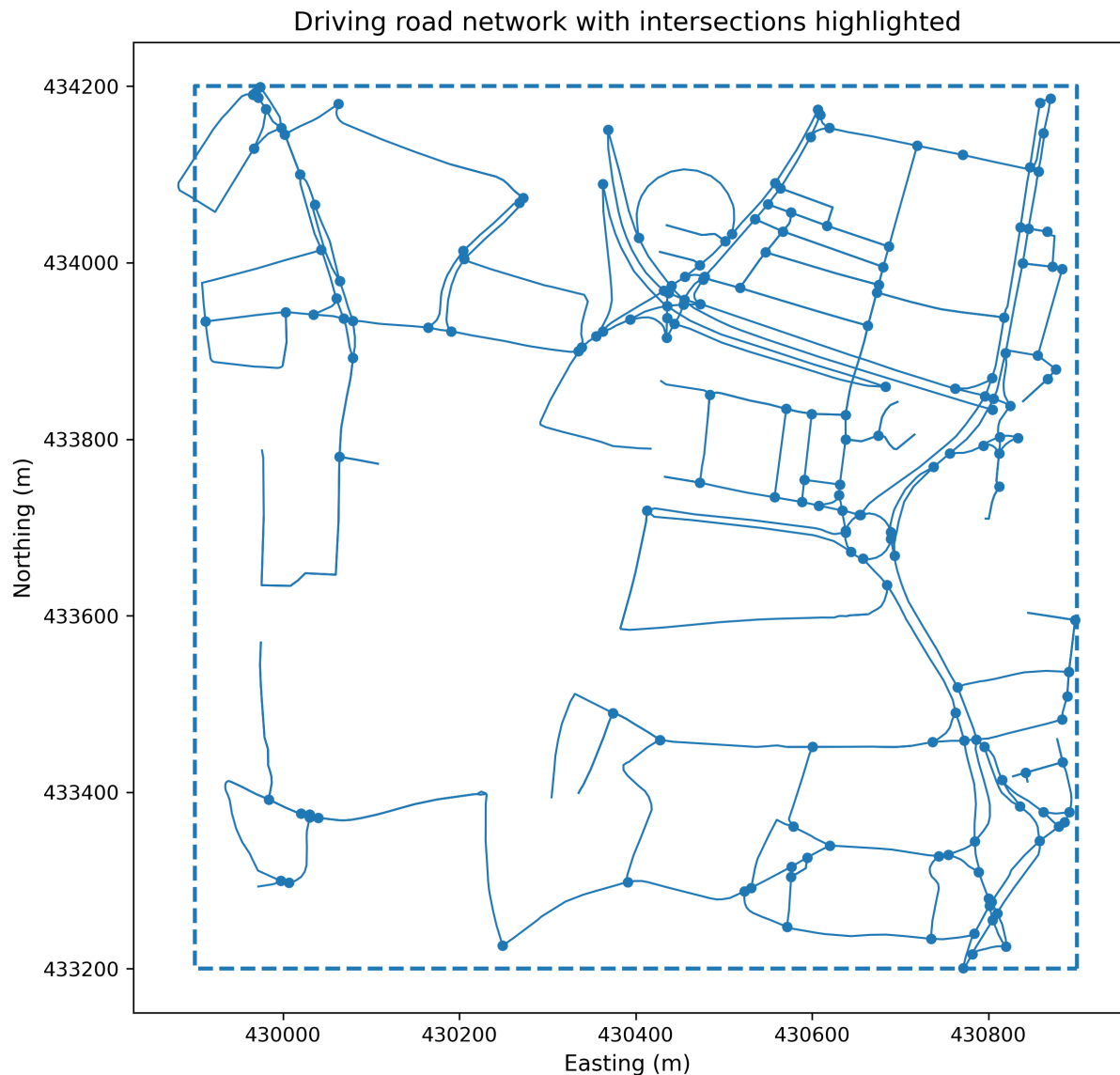


Figure 16 – Driving road network within the selected study area, with intersections highlighted. This figure emphasises the dense internal street structure and the high concentration of intersections within the selected 1 km<sup>2</sup> area.

## Part 2 – Task B (supplementary accident-network details)

Question 1: What additional details support the visualisation of accident distribution?

The Task B analysis was carried out on 452 unique accident events in the selected 1 km<sup>2</sup> central Leeds study area, aggregated across 2012–2016. The original accident locations were first snapped to the road network so that the subsequent analysis could be carried out in a network-constrained form. Figure 17a shows both the raw accident locations and the corresponding snapped positions on the road network. Its supplementary value is methodological: it makes clear that snapping does not invent a new pattern, but instead

Table 10 – Expanded summary statistics of the selected driving road network

| Metric                                      | Value    |
|---------------------------------------------|----------|
| Study area (km <sup>2</sup> )               | 1.0      |
| Total nodes                                 | 211      |
| Total edges                                 | 359      |
| Total intersections                         | 179      |
| Total edge length (km)                      | 21.834   |
| Spatial diameter (m)                        | 2930.01  |
| Average street length (m)                   | 60.82    |
| Node density (per km <sup>2</sup> )         | 211.0    |
| Intersection density (per km <sup>2</sup> ) | 179.0    |
| Edge density (m per km <sup>2</sup> )       | 21833.71 |
| Average circuitry                           | 1.6071   |
| Topologically planar                        | No       |

clarifies how the observed events align with the street system. The visual concentration of accidents along a limited number of corridors and around several major junctions is already evident before any statistical test is applied.

Figure 17b then aggregates these accidents by network arc. This panel is useful because it shifts the focus from isolated events to road segments as units of risk. The pattern is highly uneven: most arcs have zero or low totals, while a small number of streets accumulate much higher accident counts. This reinforces the conclusion that accident occurrence is concentrated on a limited subset of the network rather than being diffuse across all links.

## Question 2: What additional evidence supports the conclusion that accident risk is clustered across connected roads?

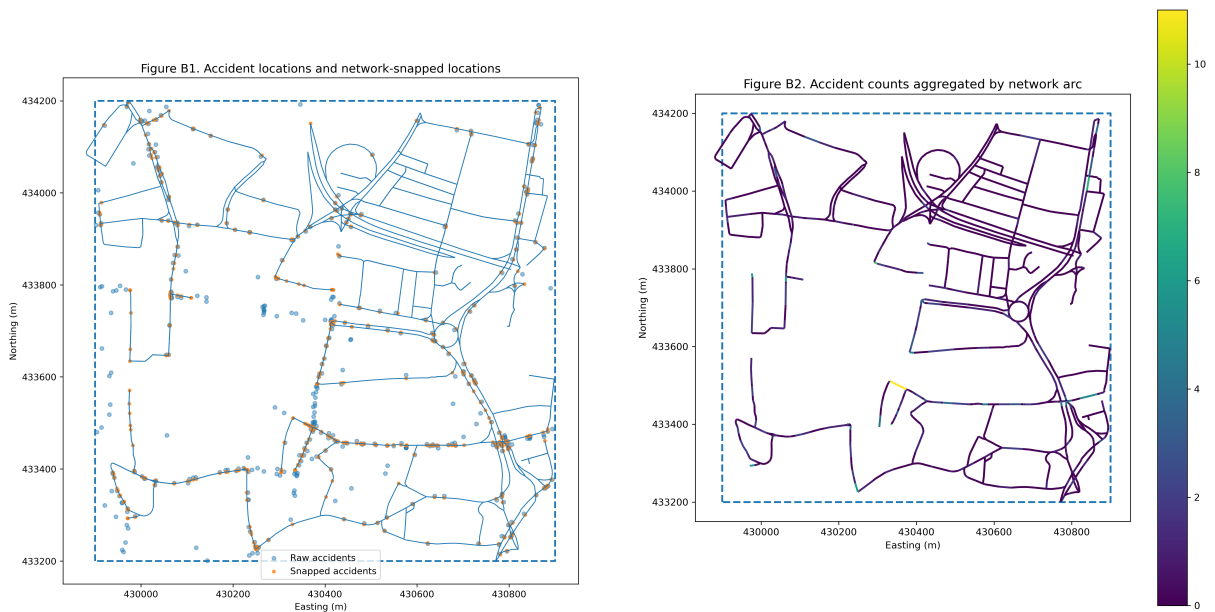
The Moran’s I analysis shows positive and statistically significant autocorrelation in accident counts on connected roads. The observed value is 0.1437, compared with an expected value of  $-0.0007$  under randomness, with  $p = 0.001$  and  $z = 5.9278$ . This means that high-accident arcs are disproportionately near other high-accident arcs, and low-accident arcs are likewise near other low-accident arcs. These summary values are reported in Table 11.

Figure 18a is the Moran scatterplot and Figure 18b is the permutation distribution. Together, these panels provide a visual check on the numerical result. The scatterplot shows a positive trend, while the permutation plot confirms that the observed Moran’s I lies well above the null range expected if accident counts were randomly arranged on the network. The global network K-function in Figure 18c provides complementary evidence by showing that the observed K-function stays above the simulated upper envelope for much

of the plotted distance range, especially beyond about 1.3 km. Considered together, these figures show that accidents cluster within broader connected zones of the road network rather than appearing as isolated independent hot spots.

Table 11 – Key summary outputs for the accident-network analysis

| Measure                                   | Value   |
|-------------------------------------------|---------|
| Unique accidents (2012–2016)              | 452     |
| Moran's $I$                               | 0.1437  |
| Expected Moran's $I$                      | -0.0007 |
| Moran permutation $p$ -value              | 0.001   |
| Moran $z$ -score                          | 5.9278  |
| Mean fraction from nearest intersection   | 0.2212  |
| Median fraction from nearest intersection | 0.1723  |
| % within nearest 10% of arc length        | 34.96   |
| % within nearest 20% of arc length        | 51.99   |
| % beyond 40% of arc length                | 21.24   |



(a) Raw and network-snapped accident locations in the selected study area. (b) Accident counts aggregated by road edge.

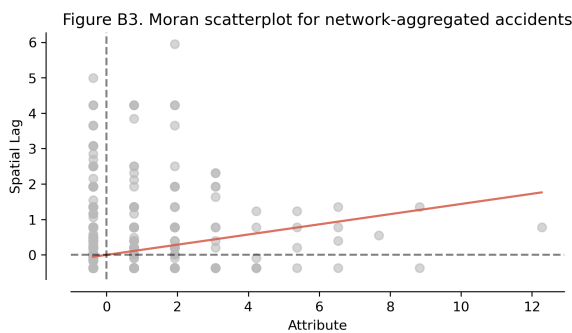
Figure 17 – Distribution of accidents on the selected road network. This figure shows that accident occurrence is not uniformly distributed, but concentrated along a limited number of roads and junction environments.

Question 3: What additional detail is useful for interpreting accident proximity to intersections?

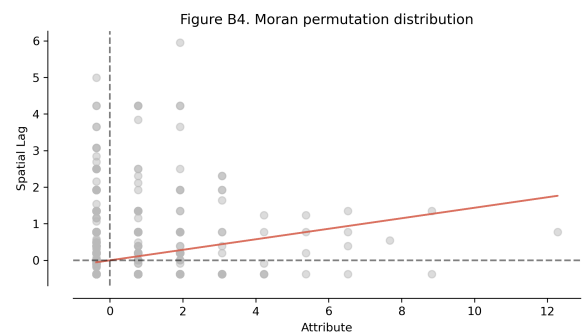
To address the third coursework question, each snapped accident was located on its host arc and expressed as a fraction of arc length from the nearest intersection. This standardises accident position relative to the geometry of each road segment, allowing

comparisons across streets of different lengths. A value near 0 indicates strong junction proximity, while a value nearer 0.5 indicates a more mid-segment location.

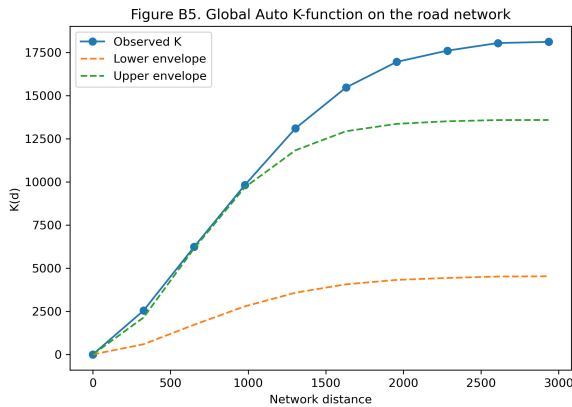
Figure 18d shows the histogram of this fraction. The distribution is strongly concentrated at small values, visually confirming that many accidents occur near intersections. The summary statistics in Table 11 are consistent with the histogram: the mean is 0.2212, the median is 0.1723, 34.96% of accidents occur within the nearest 10% of arc length from an intersection, and 51.99% occur within the nearest 20%. Only 21.24% occur beyond 40% of the road length. This pattern reinforces the interpretation that intersections and their immediate approaches are disproportionately risky parts of the network.



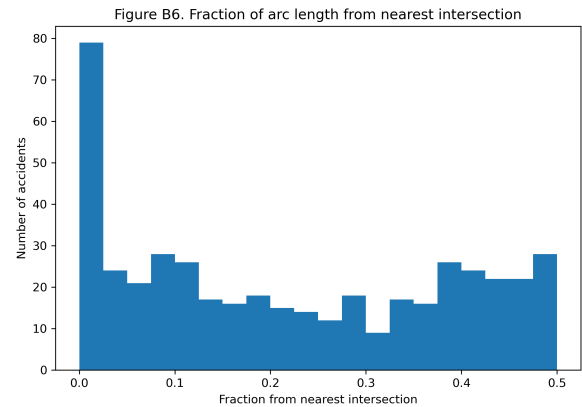
(a) Moran scatterplot for network-aggregated accident counts.



(b) Permutation distribution for Moran's I.



(c) Global network K-function with simulation envelope.



(d) Fraction of arc length from the nearest intersection for snapped accident events.

Figure 18 – Supplementary clustering and intersection-proximity diagnostics. These panels show that accident counts cluster across connected roads and that accident locations are disproportionately close to intersections.

#### Question 4: What do these results imply for road-safety interpretation?

The supplementary figures and statistics reinforce a coherent interpretation of the selected Leeds study area. Accident occurrence is neither uniform nor random. Instead, it is concentrated on a limited set of roads, clustered across connected parts of the network, and disproportionately associated with intersections and their approaches. This means

that network structure is central to understanding risk: individual dangerous links matter, but so do the connected local contexts in which those links are embedded. From a policy perspective, the results support targeted interventions at important junctions and along adjacent high-risk corridors rather than evenly distributed treatment across all streets.

## Part 2 – Task C (supplementary Voronoi and route-feasibility details)

### Question 1: What additional detail supports the seed selection and Voronoi partition?

The four initial seed points were selected using a combination of spatial separation, network accessibility, and distance from accident-prone roads identified in Task B. This strategy was intended to produce marathon cells that were not only geographically distributed across the study area, but also less exposed to the roads carrying the highest accident concentrations. As shown in Figure 19a, the selected seeds are distributed across the network rather than clustered in one local area. Figure 19b shows the corresponding node-based network Voronoi partition, in which each node is allocated to the nearest seed according to shortest-path distance along the road network.

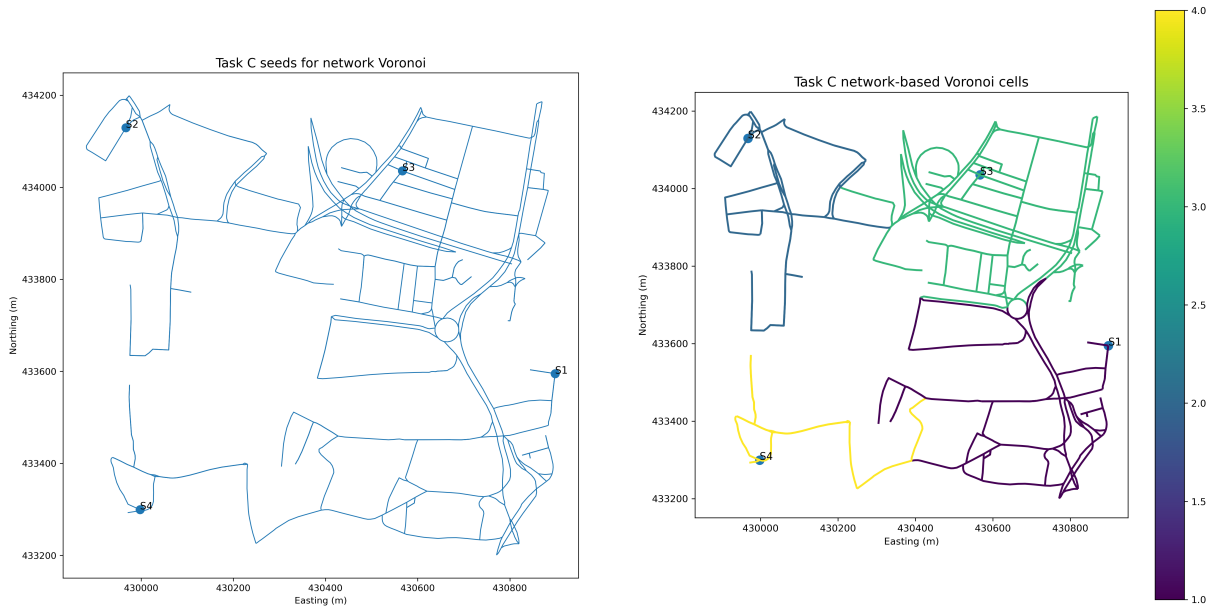
These supplementary figures are useful because they separate the two conceptual steps of the analysis: first, selecting candidate marathon starting locations; and second, transforming these locations into accessibility-based service regions. The resulting partition is spatially reasonable for accessibility analysis, but it also makes clear that the cells differ substantially in their internal network extent. This unevenness is important because it foreshadows the route-feasibility problem examined in the following questions.

### Question 2: What additional detail is useful for interpreting the initial route-search failure?

Following the coursework specification, the route-search procedure was first applied to three cells as an initial feasibility test before extending the analysis to all four cells. The supplementary results for this first stage are reported in Table 13 and visualised in

Table 12 – Demo cell summary for Task C

| Cell | Node | Edge | Road<br>length (km) | Connected | Components | Success<br>within<br>tolerance | Tested |
|------|------|------|---------------------|-----------|------------|--------------------------------|--------|
| 1    | 62   | 83   | 5.208               | 1         | 1          | 0                              | 1      |
| 2    | 32   | 43   | 3.673               | 1         | 1          | 0                              | 1      |
| 3    | 103  | 142  | 10.081              | 1         | 1          | 0                              | 1      |
| 4    | 14   | 16   | 1.582               | 1         | 1          | 0                              | 0      |



(a) Selected seed points for the network-based Voronoi partition. (b) Network-based Voronoi cells generated from the four seed points.

Figure 19 – Seed selection and network Voronoi partition. These panels show that the partition is spatially reasonable for accessibility analysis, but uneven in internal network scale.

Figure 20. The route-search procedure attempted to construct closed routes within the selected cells by combining shortest paths between sampled internal control nodes and retaining candidate routes closest to the marathon target distance of 42 km.

### Question 3: What does the redesign comparison add?

The route-search procedure was then extended from the initial three-cell test to all four Voronoi cells. The resulting four-cell summary is reported in Table 13, and the corresponding route-search outputs are shown in Figure 21. Again, no feasible closed route of approximately 42 km was found in any cell.

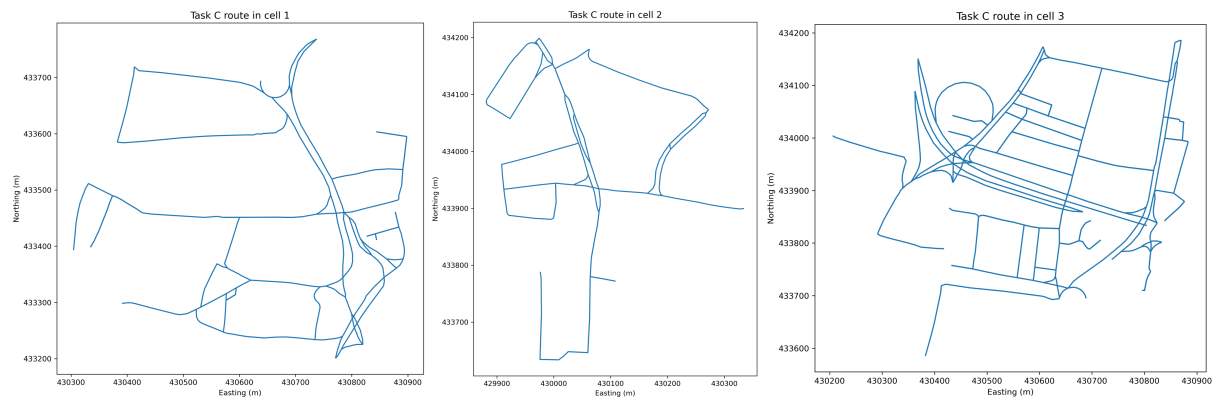


Figure 20 – Initial route-search results for the three cells tested first. None of the tested cells supports a feasible closed route close to 42 km.

Table 13 – Summary statistics of the initial four Voronoi cells and route-search feasibility.

| Cell | Node | Edge | Road<br>length (km) | Connected | Components | Success<br>within<br>tolerance | Tested |
|------|------|------|---------------------|-----------|------------|--------------------------------|--------|
| 1    | 62   | 83   | 5.208               | 1         | 1          | 0                              | 1      |
| 2    | 32   | 43   | 3.673               | 1         | 1          | 0                              | 1      |
| 3    | 103  | 142  | 10.081              | 1         | 1          | 0                              | 1      |
| 4    | 14   | 16   | 1.582               | 1         | 1          | 0                              | 1      |

The explanation is structural rather than algorithmic. As shown in Table 14, the four initial cells contain only 5.208 km, 3.673 km, 10.081 km, and 1.582 km of drivable road, respectively, and all four record `success_within_tolerance` = 0. Even allowing for repeated use of links, these totals remain far below what would normally support a meaningful closed marathon route. Cell 4 is especially restrictive, with only 14 nodes, 16 edges, and approximately 1.582 km of total road length. Figure 20 reinforces the same conclusion visually: all four cell networks are compact urban fragments rather than route-supporting marathon regions.

#### Question 4: What do these results imply for the broader interpretation of Task C?

The coursework requires a modified repeat analysis if some cells do not contain feasible routes. In this case, the selected modification was a redesign of the seed configuration intended to produce a more even partition. However, the repeated analysis led to the same substantive outcome: no feasible route of approximately 42 km was found after redesign. This comparison is reported in Table 14.

The importance of Table 14 is interpretive rather than numerical. It shows that the key bottleneck lies in the limited scale of the available road network rather than in the initial seed placement. Changing the seed configuration within the same compact 1 km<sup>2</sup> study area cannot meaningfully increase the route capacity of the cells. For that reason, the most credible future redesign would involve enlarging the study area substantially or reducing the number of marathon cells, rather than continuing to refine seed placement within the same bounded network.

Table 14 – Comparison of route-search feasibility before and after redesign.

| Cell | Initial success | Redesign success |
|------|-----------------|------------------|
| 1    | 0               | 0                |
| 2    | 0               | 0                |
| 3    | 0               | 0                |
| 4    | 0               | 0                |

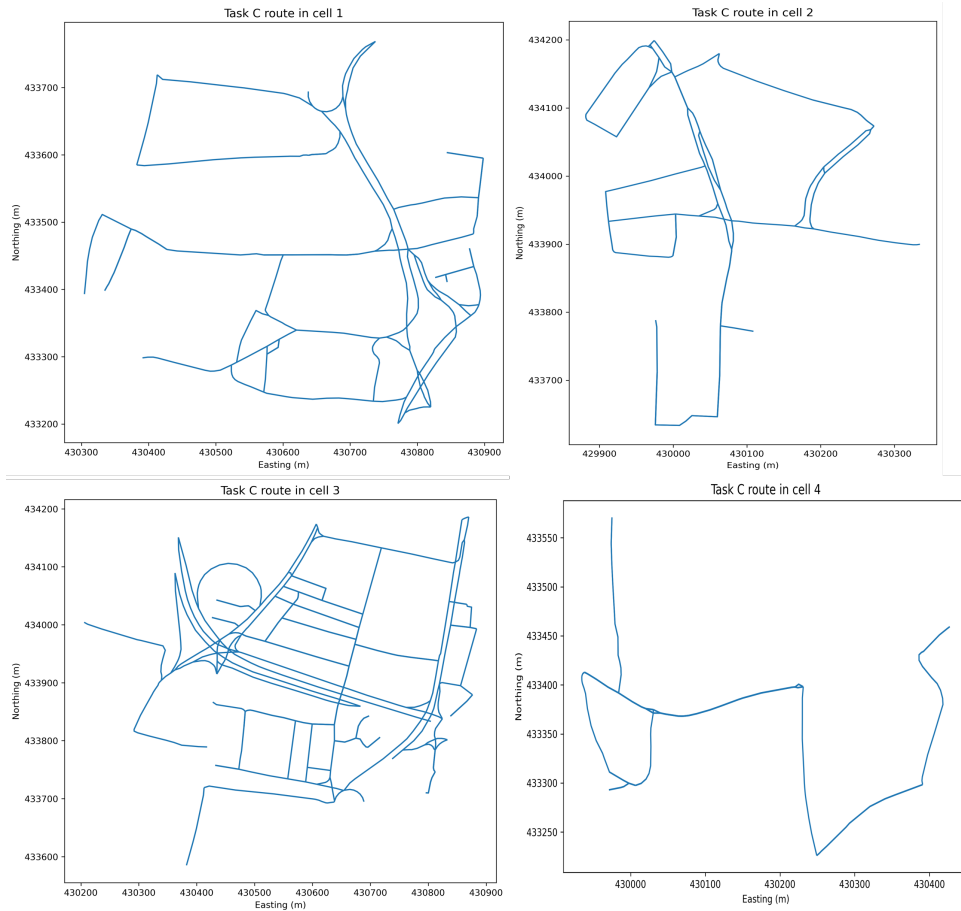


Figure 21 – Best closed-route search results by initial Voronoi cell. These panels show that all four cell networks are too limited in spatial extent to support a realistic closed route close to 42 km.

### Question 5: What do these supplementary results imply for the broader interpretation of Task C?

Taken together, Figure 19, Figure 20, Figure 21, and Tables 12–14 support the main-text conclusion of Task C. The node-based network Voronoi diagram is appropriate for dividing the selected road network into accessibility-based service regions, because accessibility should be measured along roads rather than by straight-line Euclidean distance. However, the route-search results show that a mathematically valid and visually reasonable partition is not necessarily suitable for long-distance route planning.

The broader implication is that the central Leeds study area selected for Part 2 is well suited to local road-network analysis, accessibility interpretation, and accident-risk analysis, but too small to sustain four separate marathon cells each containing a realistic 42 km closed route. In this case, the limiting factor is therefore the spatial scale of the study area rather than the Voronoi method (Okabe et al., 2000) itself.

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